HackEval AI

Multi-Step Prompt Workflow for Hackathon Idea Evaluation

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Assessment Type Mini Assessment – Prompt Workflow Design

1. Project Objective

The objective of HackEval AI is to design a structured, multi-step prompt workflow that takes a hackathon problem statement and proposed solution as input, evaluates the idea from multiple perspectives, improves its weaknesses, and produces a concise, presentation-ready final proposal. The workflow demonstrates prompt chaining, where the output of one stage is deliberately passed as context to the next stage.

2. Problem Statement

Restaurants, hotels, supermarkets, and households generate surplus food, while nearby communities and organizations may face food shortages. Lack of real-time information about available surplus food, its quantity, location, freshness, and suitable recipients can result in unnecessary food wastage. Develop an AI-powered solution that predicts surplus food, connects donors with suitable nearby organizations, and supports efficient and safe redistribution.

3. Workflow Architecture

INPUT	PROBLEM UNDERSTANDING	SOLUTION ANALYSIS	INNOVATION EVALUATION	FEASIBILITY & IMPACT	OPTIMIZE IDEA	FINAL PITCH

4. Workflow Steps and Expected Outputs

Step Workflow / Purpose Expected Output

1. Problem Understanding: Analyze the problem statement, target users, current challenges, root causes, constraints, and success criteria. Structured problem analysis containing the core problem, target users, key challenges, constraints, and desired outcome.

2. Solution Analysis: Examine the proposed solution and determine how its features address the identified problem and user needs. Solution assessment containing key features, user journey, strengths, feature-to-problem mapping, and identified gaps.

3. Innovation Evaluation: Evaluate the solution's uniqueness, innovation, differentiation, and value proposition. Innovation score (1–10), USP, differentiation opportunities, and specific improvement suggestions.

4. Feasibility & Impact :Assess technical feasibility, implementation complexity, cost, scalability, adoption potential, sustainability, and real-world impact. Feasibilityimpact scorecard with category-wise scores, major risks, and mitigation strategies.

5. Solution Optimization : Improve the original solution using the findings from the previous stages while preserving its core purpose. Optimized solution concept with improved features, innovative elements, enhanced UX, technical approach, scalability, and expected impact.

6. Final Pitch Generator Combine the validated and optimized findings into a concise, presentation-ready hackathon proposal. Final hackathon proposal containing the USP, technology, key features, impact, evaluation score, risks, and a persuasive 60-second pitch.

5. Detailed Prompt Design

Step 1 — Problem Understanding

Analyze the provided hackathon problem statement. Identify the core problem, target users or beneficiaries, current challenges, root causes, constraints, expected outcome, and key requirements that a successful solution should satisfy. Do not propose a solution yet. Return the analysis using clear headings and concise bullet points.

Step 2 — Proposed Solution Analysis

Using the problem analysis from the previous step, evaluate the proposed hackathon solution. Explain how it addresses the core problem, identify target users, main features, user journey, feature-to-problem mapping, strengths, and missing components. Do not completely redesign the solution at this stage; focus on evaluating the existing proposal.

Step 3 — Innovation & Uniqueness Evaluation

Evaluate the proposed solution for innovation and uniqueness. Identify its innovative aspects, features that are common or replaceable, unique value proposition, and opportunities for differentiation. Suggest three specific improvements that could increase novelty and provide an innovation score from 1–10 with justification. Avoid unsupported claims about competitors.

Step 4 — Feasibility & Impact Assessment

Evaluate the solution across technical feasibility, implementation complexity, cost feasibility, scalability, real-world impact, user adoption potential, and sustainability. Give each category a score from 1–10 with a concise justification. Calculate an overall score out of 10. Identify the three biggest implementation risks and one mitigation strategy for each.

Step 5 — Solution Optimization

Improve the proposed hackathon solution using the previous evaluations. Preserve the fundamental purpose while strengthening innovation, technical feasibility, scalability and user experience. Provide an improved concept, five improved features, new innovative elements, expected impact, and explain how the revised solution addresses earlier weaknesses.

Step 6 — Final Hackathon Pitch Generator

Using all previous analyses, generate the final hackathon-ready proposal. Include: project title, one-line problem statement, proposed solution, target users, key features, innovation/USP, technology approach, implementation approach, expected impact, scalability, risks and mitigations, overall evaluation score, and a persuasive 60-second pitch. Do not invent facts, statistics, partnerships or achievements.

6. Prompt Chaining & Information Flow

The workflow is designed so that each stage builds on verified context from earlier stages. Step 1 creates the problem analysis; Step 2 consumes that analysis to evaluate the solution; Steps 3 and 4 use the solution analysis to score innovation, feasibility and impact; Step 5 uses those findings to improve the concept; and Step 6 consolidates the optimized concept into the final pitch. This makes the workflow reproducible, traceable and more reliable than a single broad prompt.

7. Demonstration Example

Input Problem: Restaurants, hotels, supermarkets, and households often have surplus food that may remain unused, while nearby community organizations may need food support. Poor coordination, limited visibility of available food, uncertainty about quantity and freshness, and inefficient pickup arrangements can cause avoidable food waste. Develop an AI-powered solution that helps identify surplus food and coordinate its redistribution.

Proposed Solution: FoodBridge AI is an AI-powered platform that helps donors record or predict surplus food, identifies suitable nearby recipient organizations, prioritizes available food using factors such as quantity and freshness, and supports a coordinated pickup or delivery workflow.

Expected Final Result: After the six stages, the system should produce an evaluated and improved FoodBridge AI concept with a clear USP, technology approach, impact assessment, feasibility score, risk mitigation plan and a concise judge-facing pitch.

8. Expected Learning Outcome

This workflow demonstrates how prompt engineering can be used to decompose a complex hackathon task into manageable AI interactions. It highlights sequential prompting, contextual reuse, structured outputs, evaluation criteria, iterative improvement and final content generation—all without requiring software development or code.

9. Conclusion

HackEval AI transforms an initial hackathon concept into a structured, evaluated and pitch-ready proposal through six connected prompt stages. By separating problem understanding, solution analysis, innovation assessment, feasibility evaluation, optimization and final communication, the workflow provides a clear and professional approach to AI-assisted hackathon preparation. In the FoodBridge AI example, the same workflow can be used to systematically examine a food-waste redistribution idea, identify weaknesses, strengthen the concept and prepare a concise final pitch.

Additional Demonstration Details

- **Project Title:** FoodBridge AI – Intelligent Surplus Food Redistribution
- **Core USP:** A structured AI workflow for predicting, matching and coordinating surplus food redistribution.
- **Technology Approach:** AI/ML for prediction and matching, a web/mobile interface for donor and recipient interaction, and location-aware coordination for pickup or delivery.
- **Primary Impact:** Reduce avoidable surplus-food wastage while improving coordination between food donors and recipient organizations.
- **Overall Workflow:** Understand → Analyze → Evaluate → Assess → Optimize → Pitch.